Research Methodology Onion Model Justification

Final Year Project Activity

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Research Topic:

Design and Development of an Intelligent Emergency First Aid Guidance and Incident Response System for Untrained Users Using Human-Centred Interaction Design

1. Research Philosophy: Pragmatism

Pragmatism will be adopted as the research philosophy for this study as it focuses on practical problem-solving and supports the use of multiple research methods. This research will aim to design and develop an intelligent emergency first aid guidance and incident response system for untrained users, where both system performance and user interaction will be evaluated. Pragmatism will allow the integration of both quantitative and qualitative approaches, which is suitable for addressing real-world emergency scenarios involving human-centred interaction, usability under stress, and decision-making support.

Positivism will not be selected as it primarily emphasizes objective measurement and hypothesis testing, which does not fully capture user experience, emotional stress, and usability aspects that are critical in emergency situations. Interpretivism will also not be selected as it focuses mainly on subjective understanding and qualitative insights, which alone will not be sufficient to evaluate system effectiveness and measurable performance outcomes of the proposed solution.

Table 1: Philosophy Comparison

Philosophy	**Application to This Project**
Pragmatism (Selected)	Will support both system performance evaluation and user experience assessment using mixed methods
Positivism	Will be too focused on numerical data and will not capture usability in emergency scenarios
Interpretivism	Will focus mainly on subjective insights, which will not be sufficient alone for system evaluation

2. Research Approach: Abductive

An abductive research approach will be adopted as it allows the combination of existing theoretical knowledge with practical observations to design and refine the proposed system. This approach will be suitable because the research will build upon established emergency response guidelines and first aid knowledge while applying them in the development of an intelligent interactive system. The system will be iteratively improved based on observations, user feedback, and evaluation results.

Deductive reasoning will not be selected as it primarily involves testing predefined hypotheses, whereas this research focuses on system design and development rather than hypothesis verification. Inductive reasoning will also not be selected as it focuses on

generating new theories from observations, while this study will apply existing knowledge to develop a practical solution rather than creating entirely new theoretical frameworks.

Table 2: Approach Comparison

Approach	Application to This Project
Abductive (Selected)	Will combine existing emergency knowledge with real-world observations to refine the system
Deductive	Will be limited to hypothesis testing and not suitable for iterative system development
Inductive	Will focus on theory generation rather than applying existing emergency response knowledge

3. Methodology: Mixed Methods

A mixed-methods approach will be used in this research as it will combine both quantitative and qualitative data collection and analysis techniques. Quantitative methods will be used to measure system performance indicators such as task completion time, response accuracy, and effectiveness of guidance. Qualitative methods will be used to collect user feedback regarding usability, clarity of instructions, and overall user experience when interacting with the system during simulated emergency scenarios.

Quantitative-only methods will not be selected as they will not capture user perceptions, usability challenges, and emotional responses under stress. Qualitative-only methods will also not be selected as they will not provide measurable performance indicators required to objectively evaluate the effectiveness of the system.

Table 3: Methodology Comparison

Methodology	Application to This Project
Mixed Methods (Selected)	Will combine performance metrics and user feedback for comprehensive evaluation
Quantitative Only	Will not capture user experience and usability aspects
Qualitative Only	Will not provide measurable system performance data

4. Research Strategy: Design Science with Survey

The primary research strategy used in this study is Design Science, as the research involves the design and development of an intelligent emergency response system as an artifact. This strategy will be appropriate because the research will focus on creating a functional, user-centred solution that addresses real-world emergency response challenges identified in the research problem and background.

In addition, surveys will be used as a supporting strategy to gather user requirements and evaluate system usability from potential users. Other strategies such as experiments, case studies, simulations, and action research will not be selected as standalone strategies because they either focus on controlled environments, contextual analysis, artificial settings, or iterative cycles that do not fully support both system development and user evaluation within the scope and constraints of this project.

Table 4: Strategy Comparison

Strategy	**Application to This Project**
Design Science (Selected)	Will focus on building and evaluating the emergency guidance system as an artifact
Survey (Supporting)	Will collect user requirements and feedback
Experiments	Will focus only on controlled testing and not full system development
Case Studies	Will focus on analysis of specific cases rather than system creation
Simulation / Action Research	Will not fully cover both development and evaluation requirements

5. Time Horizon: Cross-Sectional

A cross-sectional time horizon will be adopted for this research, as data will be collected at specific points during the system development and evaluation phases. User testing and feedback will be gathered during system prototype evaluation rather than over an extended period. This approach will be suitable as the project will be conducted within the timeframe of an undergraduate final year project.

A longitudinal approach will not be selected as it requires long-term data collection and repeated observations over time, which is not feasible within the limited duration of the project.

Table 5: Time Horizon Comparison

Time Horizon	Application to This Project
Cross-Sectional (Selected)	Data will be collected once during the evaluation phase of the system
Longitudinal	Requires long-term data collection, which is not feasible within the project timeline

6. Techniques: Data collection methods

This research will use data collection methods such as questionnaires, user testing sessions, and optional interviews to gather feedback on the system. Questionnaires will be used to collect structured feedback from users regarding usability and clarity of emergency instructions. User testing sessions will be conducted where participants will interact with the system during simulated emergency scenarios to evaluate response time and accuracy. Optional interviews may be conducted to gain deeper qualitative insights into user experience.

For example, participants will be given emergency scenarios such as a heart attack or injury case, and they will follow the system's step-by-step guidance while their performance is observed.

The system will be developed using appropriate software development tools and technologies, along with UI/UX design tools to ensure a human-centred interface that is simple and intuitive under stress conditions.

Data analysis will be conducted using statistical analysis techniques to evaluate quantitative data such as response time and success rate. In addition, thematic analysis will be used to identify patterns and themes from qualitative user feedback.

Table 6: Techniques Overview

Area	Techniques Used
Data Collection	Questionnaires, User Testing, Optional Interviews
Development Tools	Software development tools and UI/UX design tools
Data Analysis	Statistical analysis (quantitative), Thematic analysis (qualitative)